

ASHOK S.

Senior AI/GenAI Engineer | Cloud & Enterprise Automation Specialist

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 LinkedIn

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AI & DevOps professional with nearly **10 years of expertise** in automation, cloud solutions, and AI-driven application development. Specialized in implementing **Generative AI, LLMs, RAG**, and **agentic AI solutions** to build production-grade AI systems. Expert in **AWS cloud platform, AWS Bedrock**, and **AWS SageMaker**, with demonstrated success in modernizing legacy systems into scalable microservices architectures. Certified in **AWS** and **Databricks AI**, with recognized expertise in prompt engineering, AI model integration, and **TypeScript/JavaScript** development.

Core Competencies

GenAI & LLM Expertise

LLM Integration, RAG Architecture, Agentic AI Solutions, Prompt Engineering, AI Model Optimization, Multi-Agent Systems

AI Frameworks & Tools

LangChain, LangGraph, Databricks AI, CosmosAI, ClaudeAI, Hugging Face

Vector Databases

Pinecone Vector DB, Chroma, FAISS, Vector Search, Embeddings Management

AWS Cloud & AI Services

AWS Bedrock, AWS SageMaker, AWS Lambda, AWS S3, AWS EC2, Serverless Architecture, Cloud Modernization

Programming & Development

TypeScript, JavaScript, Python, NodeJS, Java, Spring Boot, API Development

MLOps & Deployment

CI/CD Pipelines, Docker, Kubernetes, Jenkins, Terraform, Production Monitoring, AI Governance

Awards & Recognition

Spirit of Chennai Award

Agile Imaginative Award for Value-Added Deliverables

Key Accomplishments

- ★ Designed and implemented **AI-assisted dashboards**, reducing monitoring blind spots by **70%**
- ★ Reduced repetitive data processing efforts by **40%** using **GenAI-powered automation solutions**
- ★ Enhanced application error detection accuracy by **75%** using **AI-driven anomaly detection tools**
- ★ Improved cloud resource utilization efficiency by **20%** through serverless and containerized deployments
- ★ Accelerated monolith-to-microservices migration, improving deployment predictability by **30%**
- ★ Optimized cross-functional team workflows, resulting in **20%** improvement in collaborative project delivery efficiency

Technical Skills

Programming Languages

TypeScript, JavaScript, Python, NodeJS, Java, Shell Scripting, SQL, PL/SQL

AI/ML Frameworks

LangChain, LangGraph, Databricks AI, CosmosAI, ClaudeAI, Hugging Face, PyTorch

Vector Databases

Pinecone, Chroma, FAISS, Qdrant

AWS Cloud Services

AWS Bedrock, AWS SageMaker, AWS Lambda, AWS S3, AWS EC2, AWS CodePipeline

DevOps & CI/CD

Jenkins, Git, Terraform, Docker, Kubernetes, Artifactory, Pipeline Automation

Monitoring Tools

Grafana, Prometheus, Kibana, MLOps Monitoring

Education & Certifications

MCA – Anna University

B.Sc. Computer Science – Bishop Heber College

AWS Certified Cloud Practitioner

AWS Knowledge: Serverless

AWS Well-Architected Proficient

Databricks – Generative AI

Databricks – AI Agent

Databricks – AI Security

Diploma in Innovation Management – IBM Institute, Berlin

Diploma in Sustainable Management – IBM Institute, Berlin

DevOps Engineer 3 | ADITI

Oct 2024 – Jul 2025 | Project: PayPal (Payment & Instrument Vault)

- Architected and deployed **production-grade AI chatbot** leveraging **LLMs**, **RAG architecture**, **LangChain**, and **CosmosAI** for PayPal Funding Instrument Platform, processing 10K+ queries with **95% accuracy**
- Built **multi-agent AI systems** using **LangGraph** and **ClaudeAI** to orchestrate complex workflows, automating payment instrument validation and fraud detection pipelines
- Implemented **vector database solutions** using **Pinecone** for semantic search and document retrieval, enabling **sub-second query responses** on 100K+ financial documents
- Developed **TypeScript/JavaScript APIs** integrating **LangChain agents** with backend microservices, enabling real-time AI-powered decision making in payment workflows
- Engineered **AWS Bedrock** integration for foundation model deployment, optimizing inference costs by **35%** while maintaining performance SLAs
- Built **MLOps pipelines** using **AWS SageMaker** and **Docker** for continuous model training, evaluation, and deployment with automated rollback capabilities
- Designed comprehensive **AI observability dashboards** in **Grafana** tracking model performance metrics, token usage, latency, and accuracy across production environments
- Implemented **prompt engineering frameworks** with version control and A/B testing, improving response quality by **45%** and reducing hallucinations
- Automated recurring production tasks using **GenAI-powered scripts**, **reducing manual operational efforts by 40%** and improving team efficiency
- Established **AI governance and security protocols**, including PII detection, content filtering, and compliance monitoring for financial regulations

Senior Software Engineer | AgileAxis Pvt. Ltd

Sep 2022 – Sep 2024 | Project: AthenaHealth (Rules 2.0 Migration)

- Led development of **AI-powered healthcare rules engine** using **LangChain** and **LLMs** to automate medical billing validation, reducing processing time by **60%**
- Architected **RAG-based system** for medical code classification, integrating **vector databases (Chroma)** with healthcare knowledge bases to achieve **92% accuracy**
- Built **intelligent document processing pipeline** using **GenAI models** to extract and classify medical claims data from unstructured documents, processing **50K+ documents daily**
- Developed **TypeScript/NodeJS microservices** integrating **AI agents** with legacy healthcare systems, enabling real-time clinical decision support
- Implemented **AWS Lambda** and **AWS Bedrock** serverless architecture for AI model inference, achieving **99.9% uptime** and handling **1M+ daily requests**
- Created **prompt templates library** and optimization framework, standardizing AI interactions across 15+ healthcare applications
- Designed **MLOps workflows** with automated testing, model versioning, and deployment using **CI/CD pipelines** and **Docker containers**
- Led team of 8 engineers in migrating monolithic healthcare application to **microservices architecture**, improving scalability and deployment frequency by **10x**
- Established **AI model monitoring** and performance tracking dashboards, enabling proactive identification and resolution of model drift
- Conducted rigorous code reviews and implemented **AI governance best practices**, ensuring HIPAA compliance and data privacy

Support Lead | Standard Chartered

Jan 2021 – Sep 2022 | Project: ICDD (Integrated Client Due Diligence)

- Automated application release and deployment processes using **Jenkins and Rundeck**, increasing operational efficiency
- Managed load-balanced environments for multiple applications, enhancing system availability and resilience
- Conducted source code audits to maintain high quality and robust application performance
- Streamlined deployment workflows, reducing errors and manual intervention

Senior Support Analyst | BNP Paribas

Jan 2020 – Jan 2021 | Project: Vision Client Collateral Management

- Acted as SME and change coordinator for L3 application support, ensuring smooth release and deployment cycles
- Automated repetitive tasks through Shell scripting and Oracle PL/SQL, reducing manual efforts and improving SLA adherence
- Mentored and trained graduate resources in L1/L2 support, enhancing team productivity

Senior Support Engineer | HCL

Dec 2017 – Jan 2020 | Client: BNP Paribas | Projects: Money Transfer Management, Cash Claim Management, MTS Swift Tool

- Delivered L2/L3 support for money transfer systems, ensuring consistent service uptime and incident resolution
- Developed **automation scripts using Shell and Oracle PL/SQL**, reducing manual operational efforts
- Managed incidents and post-incident processes, conducting root cause analysis to prevent recurrence
- Coordinated application release and deployment activities, ensuring high availability and compliance
- Supervised a 12-member team, maintaining seamless operations and meeting SLA targets
- Successfully managed Cash Claim Management sub-application, improving client claim processing efficiency

Software Engineer | Syntel

Sep 2014 – Dec 2017 | Clients: State Street Investment Management Services, FedEx

- Developed and maintained **enterprise data warehouse operations**, ensuring accurate master, market, and reference data management
- Automated financial data loads and reporting processes, reducing processing time and improving operational efficiency
- Implemented cross-functional collaboration to streamline data integration and ensure seamless project delivery
- Designed and executed **hybrid automation testing frameworks using Selenium and Java**, improving testing coverage and efficiency
- Automated test environment setups and execution, reducing manual interventions and enhancing reliability
- Created dynamic object-learning tools and performance analysis reports, supporting informed business decision-making

Open Source AI Projects & Portfolio

Portfolio Website: <https://www.profolia.art/Ashok>

Synth AI

<https://solutionsynth.cloud/>

Comprehensive AI solutions platform powered by **AWS services** and modern AI frameworks.

CarbonLens AI

<https://carbonlens-ai.solutionsynth.cloud/>

Intelligent supply chain carbon footprint tracker powered by **AWS AI services**. Automatically processes logistics documents and provides optimization recommendations using **AI/ML models**.

AI Resume & Cover Letter Generator

<https://resume-generator-ai.solutionsynth.cloud/>

Intelligent career document generation platform powered by **AI**. Creates professional, ATS-optimized resumes and personalized, compelling cover letters tailored to job descriptions using **LLMs, natural language processing**, and advanced **GenAI** with **prompt engineering techniques**.